

Aparnaa Senthilnathan

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OBJECTIVE

Recent B.S./M.S. graduate from RIT with internship and research experience across applied AI, computer vision, model evaluation, and software development. Built multi-agent AI systems, audio ML applications, and full-stack software, with a focus on putting AI techniques to good use and turning them into usable software tools.

EDUCATION AND CERTIFICATIONS

Rochester Institute of Technology

Bachelor of Science & Master of Science in Computer Science, RIT Presidential Scholar

Rochester, NY

August 2021 – May 2026

Certifications:

IBM SkillsBuild Data Analytics Certificate – IBM, 2026

Google AI Essentials – Google, 2025

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, SQL

AI/ML: PyTorch, scikit-learn, OpenCV, RAG, LLM Agents, Computer Vision, Audio ML, Model Evaluation

Development: FastAPI, React, Node.js, REST APIs, Docker

Tools: Git, Ollama, Slurm, TensorBoard

EXPERIENCE

Computer Vision Intern

May 2024 – August 2024

Kitware, Inc.

Carrboro, NC

- Trained and evaluated YOLOX and RT-DETR object-detection pipelines on around 20,000 TinyBirds training and validation images to study model performance on small-object detection
- Implemented and compared two object-detection pipelines, using PyTorch, TensorBoard, Slurm, and a multi-GPU environment to analyze training behavior and debug dataset and model-configuration issues
- Evaluated model predictions against ground-truth annotations to identify detection errors and assess how well the models recognized very small objects

Artificial Intelligence Research Co-Op

July 2023 – December 2023

Griffiss Institute

Rome, NY

- Designed experiments to evaluate how CLIP vision-language foundational models handled 20 homonym image samples, testing their ability to distinguish visually different objects sharing the same name
- Compared two CLIP zero-shot classifiers—ResNet50 and ViT-L/14—across two levels of granularity, using cosine similarity, NumPy, and confusion matrices to analyze image-image and image-text relationships
- Observed that the models produced stronger image-to-image matching than image-to-text matching, while the two CLIP classifiers showed similar overall behavior, highlighting difficulty with ambiguous visual concepts

PROJECTS

MAARA: Multi-Agent Automated Repair & Refinement Assistant | *Team Project*

April 2026 – May 2026

- Co-developed a five-agent LLM software-repair pipeline that decomposes coding into planning, validation, RAG-grounded generation, critique, and iterative repair, with models served locally through Ollama
- Evaluated heterogeneous and single-model agent configurations, with the strongest reported few-shot setup reaching 100% pass@1 and showing a 13-point gain from the strategy/evaluator gate

Mood-y: Voice-Based Mood Music Recommender | *Team Project*

March 2026 – May 2026

- Co-developed an end-to-end system that turns voice input into personalized music recommendations, extracting 112-dimensional acoustic features with Librosa and connecting emotion classification to a FastAPI/React application
- Trained and compared classifiers across eight emotions using RAVDESS audio, with the best SVM configuration achieving 91.7% accuracy and 0.916 F1 while revealing emotion-specific classification errors

FrequencyPrint: An Automated Deepfake Detection System | *Capstone Project*

January 2026 – May 2026

- Developed an audio-deepfake detection system that compares three CNN models—SimpleCNN, ResNet18, and ResNet34—across real and AI-generated music using PyTorch, Librosa, and spectrogram analysis
- Identified severe dataset overfitting after models reporting 99-100% accuracy failed on AI-generated covers, then redesigned the experiment with real and deepfake-cover data and achieved a more realistic best accuracy of 61.5%